

Middleware Lab: Connection Pool Capacity Design

Duration : 45 Min.

Scenario

During user logins, applications throw database connection acquisition timeouts. Connection pooling limits must be optimized.

Task Objectives

Perform the following actions inside the `student_workspace` directory: - Modify DataSource settings inside `context.xml` to prevent connection exhaustion and database locks.

Instructions to Perform the Task

1. When your workspace loads in **VS Code**, use the **Explorer** panel on the left to locate your files.
2. Navigate to and click the specific files mentioned in the Task Objectives.
3. Make the necessary code edits in the editor.
4. Press `Ctrl + S` (Windows) or `Cmd + S` (Mac) to save your changes.
5. If you need to run commands, open the built-in terminal by clicking **Terminal > New Terminal** from the top menu.

Validation

Once you have saved your files and are ready to submit, return to the platform dashboard and click the **"Submit Final"** button. This task is evaluated manually by the grading team.

Evaluation Rubric:

1. Configuring maximum active and idle database connections: 3 marks
2. Setting connection wait thresholds to prevent thread starvation: 3 marks
3. Configuring connection validation queries (`testOnBorrow`): 4 marks

Total Score: 10 Marks

Important Notes

- This is a manually evaluated question.
- Ensure all logs and configurations are fully saved in your workspace folder before submitting.